QP CODE

Reg. No:
Name:

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION March 2026
Fourth Semester
Discipline Specific Elective (DSE) - MG4DSEBTG200 - BIOSAFETY AND BIOETHICS
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)
Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One word or sentence
Answer any 10 questions.
Each question carries 2 marks

1. Mention one standard procedure involved in Good Laboratory Practices (GLP). [U] / [CO2]
2. What is the primary purpose of a Standard Operating Procedure (SOP) in microbe research? [U] / [CO2]
3. Name one national or international regulation governing biosafety. [K] / [CO3]
4. Define the term "Biosafety." [U] / [CO1]
5. What does the acronym PPE stand for in a laboratory context? [K] / [CO4]
6. Define "Risk Assessment" as it relates to biological research. [K] / [CO4]
7. Mention one ethical requirement for conducting research on experimental animals. [U] / [CO5]
8. Write one fundamental right of a participant in a clinical trial. [U] / [CO5]
9. State the primary function of a Biosafety Cabinet. [U] / [CO4]
10. How many minutes (minimum) should eyes be flushed with water following exposure?. [U] / [CO5]
11. List one specific design feature required for a containment laboratory. [U] / [CO3]
12. Define “biohazard”. [K] / [CO4]

(2x10= 20)

Part B

Short essay
Answer any 4 questions
Each question carries 5 marks

13. Describe the general procedures and importance of maintaining good laboratory practices. [U]/[CO2]
14. Compare and contrast the different levels of Biosafety containment (BSL 1-4). [A] / [CO3]
15. Explain different types of written communications in science. [U] / [CO3]

16. Explain the importance of hazard identification and evaluation in risk management. [U] / [CO4]
17. Discuss the history and origin of Bioethics as a scientific discipline. [U] / [CO5]
18. Outline the ethical and legal norms required for conducting preclinical studies. [K] / [CO5]

(5x4=20)

Part C

Essay type Questions

Answer any 2 questions.

Each question carries 15 marks

19. Provide a comprehensive account of the **design of containment facilities** and how they justify the various biosafety levels. [U] / [CO3]
20. Evaluate the **risk assessment and management process**, including the role of emergency response plans in biological research. [E] / [CO4]
21. Discuss in detail the **ethical and legal aspects** of working with experimental animals and plants, ensuring adherence to international standards. [An] / [CO5]
22. Explain the **legal requirements and ethical norms** for conducting clinical trials, with a focus on protecting the rights of human participants. [U] / [CO5]

(15X2=30)